



Effects of manifold design parameters on flow uniformity in parallel mini-channels

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ABSTRACT

Flow maldistribution in parallel mini-channels is a critical obstacle to the efficient design of mini-channel heat sinks. The flow maldistribution is known to cause dryout incipience in some channels, which triggers critical heat flux, eventually leading to physical burnout and catastrophic device failure. Hence, enhancing flow uniformity by reducing flow maldistribution is critical. While several attempts have been made to reduce flow maldistribution, most have focused on providing strategic frameworks for enhancing flow uniformity in channels. The present study develops a new predictive tool for quantitatively assessing flow maldistribution in parallel channels based on the parametric effect. Five parameters are considered: channel height, channel width, header length, connection tube diameter, and total volumetric flow rate. Thirty-six different heat sinks are numerically investigated to study the influence of each parameter on flow uniformity in channels. A decrease in channel height, channel width, and total volumetric flow rate and an increase in header length and connection tube diameter reduces the maldistribution factor enhancing channel flow uniformity. By analyzing the effects of these parameters on the flow uniformity in channels, a new correlation for the maldistribution factor constructed with dimensionless groups, including the design parameters, is proposed, which fits the numerical predictions for the maldistribution factor with excellent accuracy, evidenced by a root-mean-squared-error of 1.0%. The parametric assessment of the effects of the different parameters on the flow maldistribution using the new correlation shows that the magnitude of the relative influence of each variable on flow maldistribution can be expressed in the following order: header length > channel height > connection tube diameter > total volumetric flow rate > channel width.

1. Introduction

Recent advances in modern applications such as aerospace, avionics, radars, defense systems, electric vehicles, and computer data centers have driven an increasing demand for thermal management solutions to remove high heat fluxes from small surface areas [1,2]. Air-cooling heat sinks attempted to tackle the additional increase in heat load using more powerful fans or enhanced fin structures. However, due to the continuous escalation of heat dissipation rates, interest has shifted to exploring two-phase liquid-cooling schemes as the new cooling solutions. While a variety of two-phase liquid-cooling schemes including natural convection, channel flow, jet impingement, and spray cooling have been the subject of intense interest, remarkable attention has been placed on mini/micro-channel heat sinks because of their unique attributes, including superior heat removal capability, compact dimensions, and

reduced cooling fluid inventory [1,2].

Spurred by the advantages of mini/micro-channel heat sinks, extensive efforts have been made to acquire a fundamental understanding of various aspects of complex two-phase transport behaviors specifically intended for mini/micro-channel flows [3–9]. For example, several studies were undertaken to construct a universal flow pattern map to accurately predict flow patterns and transitions among two-phase flow regimes in mini/micro-channel [3–5]. Another topic of recent studies has been to assess the influence of body force on two-phase pressure drop and heat transfer and to establish criteria for negating gravitational effects by conducting flow boiling and condensation experiments in various platforms such as parabolic flight and the International Space Station (ISS) [6–7]. A few experiments concerning two-phase flow heat transfer have been conducted to provide practical means of eliminating severe flow instabilities by examining the trigger mechanisms for different types of instability in two-phase systems [8–9].

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Nomenclature

$c_1 - c_{11}$	constants in Eq. (16)	Re_{in}	Reynolds number in inlet connection tube
C_1, C_2	empirical constants in turbulent dissipation rate equation	$RMSE$	root-mean-squared-error
d	grid size; lateral distance between inlet/outlet connection tube defined in [30], (m)	S	modulus of mean rate-of-strain tensor
D_f	hydraulic diameter of channel, $D_f = 2H_{ch}W_{ch}/(H_{ch} + W_{ch})$, (m)	U	time-averaged velocity, (m/s)
D_{in}	connection tube diameter, (m)	U_{in}	velocity in inlet connection tube, (m/s)
f	friction factor	$\bar{U}$	average velocity in channel, (m/s)
G	dimensionless geometric parameter defined in [30]	V_{in}	total volumetric flow rate in inlet connection tube, (m ³ /s)
G_k	production of turbulent energy	W_{ch}	channel width, (m)
H_{ch}	channel height, (m)	W_f	fin width, (m)
k	turbulent kinetic energy, (J/kg); channel number	W_h	header width, (m)
L_{ch}	channel length, (m)	x, y, z	spatial coordinates
L_h	header length, (m)		
MAE	mean absolute error	Greek symbols	
MF	maldistribution factor	β	channel aspect ratio
n	number of numerical data for maldistribution factor	γ	maximum mass flow rate variation defined in [31], (kg/s)
N	number of channel	ε	turbulent dissipation rate, (J/kg•s)
P	pressure, (Pa)	η	strain rate, s ⁻¹
$\Delta P_{ch,avg}$	channel pressure drop averaged over entire mini-channels, (Pa)	μ	dynamic viscosity, (kg/m•s)
ΔP_{ch}	channel pressure drop, (Pa)	μ_t	turbulent viscosity, (kg/m•s)
ΔP_{tot}	total pressure drop between inlet and outlet connection tubes, (Pa)	ν	kinematic viscosity, (m ² /s)
Q_{avg}	mass flow rate averaged over entire mini-channels, (kg/s)	ρ	density, (kg/m ³)
Q_k	mass flow rate in kth channel, (kg/s)	σ_ε	empirical constant in turbulent dissipation rate equation
Q_{max}	maximum mass flow rate in channels, (kg/s)	σ_k	empirical constant in turbulent kinetic energy equation
Q_{min}	minimum mass flow rate in channels, (kg/s)		
Re	channel Reynolds number	Subscripts	
$Re_{ch,avg}$	channel Reynolds number averaged over entire mini-channels	app	apparent
		$comp$	computed
		d	fully-developed
		in	inlet
		i, j	direction indices
		$pred$	predicted

Despite the critical knowledge gained from these prior efforts, only limited information concerning reliable heat sink design has been available in the literature. Hence, several critical aspects of two-phase flow physics in mini/micro-channels were explored for better design reliability and safety of mini/micro-channel heat sinks [10–18]. To provide a foundation for a comprehensive understanding of annular flows, which generally contribute to high heat transfer coefficients in most mini/micro-channel flow systems, theoretical models for annular flow boiling [10] and condensation [11] were constructed using a control volume approach capable of accounting for various effects such as interfacial dampening of turbulent eddies, droplet entrainment, and deposition. In pursuit of system optimization and effective thermal design of mini/micro-channel heat sinks, universal predictive methods for key design parameters that apply to broad ranges of operating conditions and working fluids were developed based on consolidated databases amassed from numerous sources [12–16]. These parameters include two-phase pressure drop [12,13], two-phase heat transfer coefficient [14,15], and dryout incipience quality [16]. Important heat transfer and flow limits of mini/micro-channel heat sinks, including premature critical heat flux (CHF), dryout incipience, and two-phase critical flow [17], were examined, and relationships among two-phase critical flow, CHF, and premature CHF in micro-channel heat sink [18] were assessed.

One of the most practical issues associated with mini/micro-channel heat sinks is minimizing flow maldistribution in parallel channels. The thermal design of mini/micro-channel heat sinks is generally based on the assumption that fluid is uniformly distributed to an individual channel. In practice, however, uniform flow distribution in multiple parallel channels is not necessarily ensured. Therefore, theoretical and empirical design methods can fail to produce reliable estimates of

temperature rise in heat sink base, pressure drop, and heat transfer coefficient in channels for the safe design and operational limits of mini/micro-channel heat sinks [19]. Further, the flow maldistribution can lead to high residual stresses, severe warping of heat sink material, and large spatial temperature gradients [20]. Moreover, for mini/micro-channel heat sinks specialized in phase-change thermal management systems suited for high heat density applications, flow maldistribution is especially problematic due to the increased likelihood of causing dryout incipience and critical heat flux (CHF). The dryout incipience occurs in some channels when the partial annular liquid film begins to dry out, and the CHF can be triggered by vapor blanket formation along channel walls; both are known to be a primary cause of physical burnout and catastrophic device failure [18,21].

Mueller and Chiou [22] segregated the types of flow maldistribution in a manifold system based on several design factors, including thermophysical properties of fluid, two-phase flow instability, manufacturing tolerances, header design, and inlet/outlet arrangement. Among these design factors, header design such as shape and orientation of headers and inlet/outlet arrangement has been the focus of extensive research because they account for most cases of flow maldistribution observed in single-phase liquid-cooling heat sinks. Fig. 1 shows the schematic of typical header shapes used in mini-channel heat sinks, and Fig. 2 shows the different inlet/outlet arrangements typically used in mini-channel heat sinks. Note that the naming of the inlet/outlet header arrangements varies with authors. Hence, it is recommended to refer to the particular study to understand the inlet/outlet header arrangements. However, Fig. 2 represents the typical method of naming the inlet/outlet header arrangements. Xia et al. [23] performed numerical simulations to examine the effect of different inlet/outlet arrangements (C-, Z-, and I-type) and inlet/outlet header shapes (rectangular, symmetry

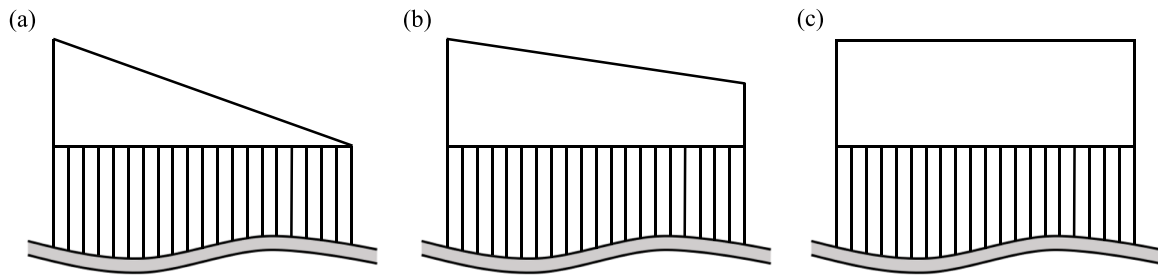


Fig. 1. Schematic of typical header shapes used in inlet/outlet of mini-channel heat sinks. (a) Triangular header, (b) trapezoidal header, and (c) rectangular header.

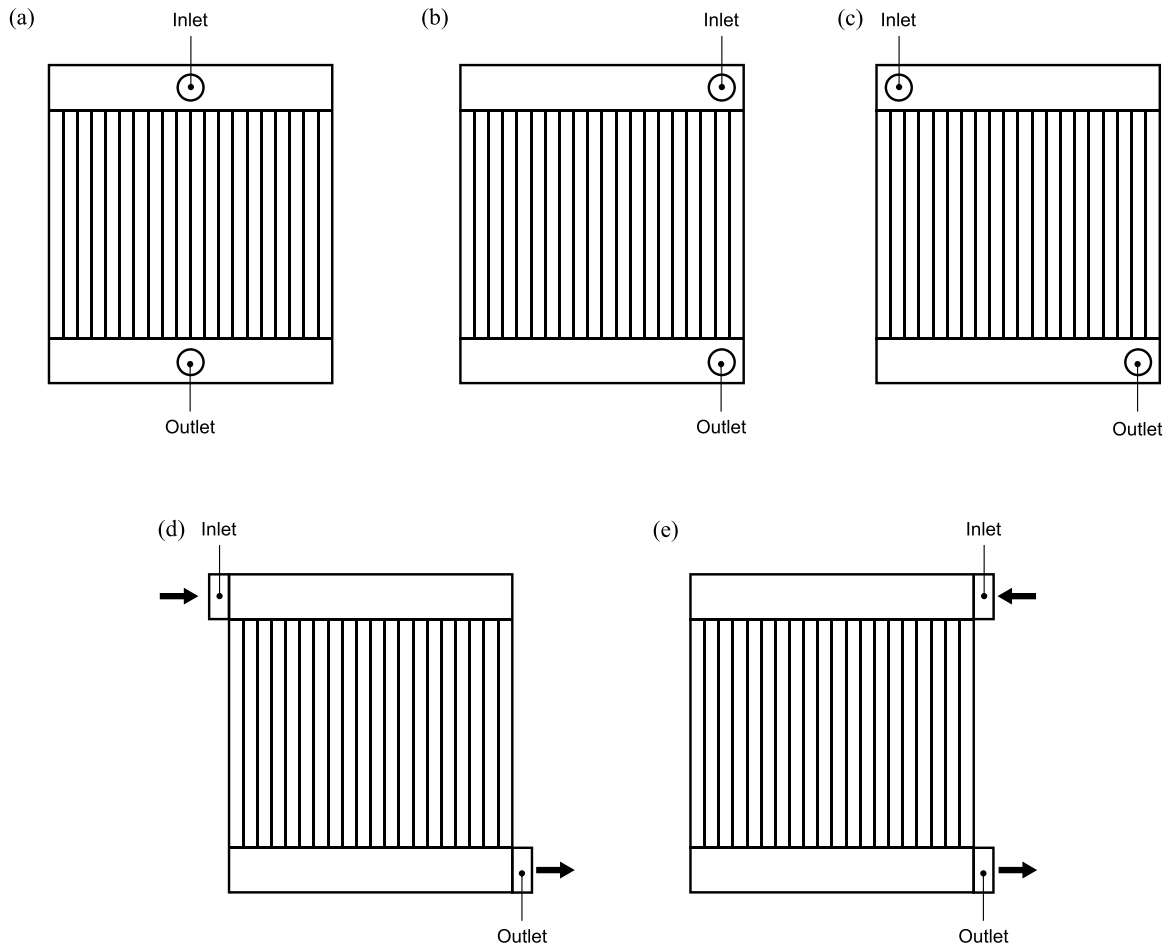


Fig. 2. Schematic of mini-channel heat sink with different inlet/outlet arrangements. (a) I-type, (b) C-type, (c) V-type, (d) Z-type, and (e) U-type.

trapezoidal, and symmetry triangular) on flow distribution. The results revealed that a comparatively uniform flow distribution was obtained from the I-type, and the C- and Z-type provided poor flow uniformity in channels due to the relatively dominant flow separation in the inlet header. Based on the examination on inlet/outlet arrangement, the I-type was selected for their subsequent study on inlet/outlet headers, and the result indicated that the rectangular header shape was advantageous compared to the other header shapes. Anbumeenakshi and Thansekhar [24] performed experiments to examine the effect of inlet header shape, inlet flow arrangement, and total flow rate on flow distribution in channels. Three different shapes of inlet header (rectangular, trapezoidal, and triangular) and two different flow inlet arrangements (horizontal and vertical inlet) were investigated. Horizontal flow inlet refers to the horizontal inlet arrangement where the fluid enters the inlet header horizontally, while the vertical flow inlet arrangement refers to the fluid entering the inlet header vertically. The

results showed that the rectangular header exhibited the worst flow uniformity in channels at lower total flow rates. On the other hand, at higher total flow rates, the most uniform flow distribution was achieved with the rectangular header since intensified vortex at the entrance of the trapezoidal and triangular headers affected the flow distribution. An experimental and numerical study was conducted by Kumaran et al. [25] to suggest the key design recommendations to enhance flow uniformity by taking into account different inlet/outlet header shapes (rectangular, trapezoidal, and triangular), header sizes, and inlet/outlet arrangements (I-, C-, V-, Z-, and U-type). Proposed design recommendations for enhanced flow uniformity in channels are the combination of C-type inlet/outlet arrangement, triangular shape for inlet header, and trapezoidal shape for outlet header. Tong et al. [26] conducted numerical simulations to identify various geometric parameters which affect flow distribution in parallel channels, such as header height-to-channel width ratio, diameter of inlet connection tube, and

cross-sectional area of inlet header. The results revealed that a better flow uniformity in channels could be obtained by increasing the header height, the diameter of the inlet connection tube, and the cross-sectional area of the inlet header. Chein and Chen [27] numerically examined the effect of different inlet/outlet arrangements (I-, D-, N-, S-, U-, and V-types) on flow maldistribution. The results showed that the I-, D-, N-, and S-types having horizontal fluid supply and collection provided relatively severe flow maldistribution in channels, attributed to flow recirculation in the inlet header. Flow uniformity in channels was significantly enhanced for U- and V-types having vertical inlet/outlet arrangements due to the flow impingement effect. Numerical simulations were performed by Mishan et al. [28] to investigate the effect of three different header designs (rectangular, straight-convergent, and divergent-straight-convergent shapes) on flow maldistribution. The results revealed that the conventional rectangular header exhibited a poor flow uniformity in channels, while enhanced flow distribution was obtained from the straight-convergent header.

Recent studies focused on several approaches to improving the flow distribution in channels. Said et al. [29] numerically studied two types of inlet header designs by introducing an orifice and a nozzle. The authors found that using the orifice reduced the flow maldistribution by 12 times with a 7.8% increase in overall pressure drop. Similarly, the inlet header with the nozzle observed a 7.5 times decrease in flow maldistribution with a 9.8% increase in overall pressure drop. Wei et al. [30] used a perforated baffle in the inlet header and developed a Computational Fluid Dynamics (CFD)-based algorithm to optimize the topology of the baffle to obtain target flow distribution in parallel channels. Later, Wei et al. [31] performed CFD simulations and experimental Particle Image Velocimetry (PIV) measurements to validate the baffle optimization algorithm. Ashrafi et al. [32] designed a novel Z-type parallel flow-field configuration for proton exchange membrane fuel cells to overcome the flow maldistribution problem in the conventional Z-type parallel flow-field configuration. Kumar et al. [33] proposed a novel approach of variable width parallel channels to reduce the flow maldistribution by 93.17% compared to a conventional design. Madane and Kulkarni [34] reduced the flow maldistribution by introducing pressure equalization slots across the parallel channels at an equal distance from the inlet and outlet of the channels. Pistoresi et al. [35] improved the flow uniformity in a multi-scale fluidic network with 90 parallel mini-channels by using geometrically optimized perforated baffles. Yadav et al. [36] mitigated flow maldistribution by splitting the single inlet port in the inlet header into two inlet ports. Moreover, they tested two different positions of the two inlet ports; at the front and on each side of the inlet header, which reduced 26.2% and 68.5% of flow maldistribution, respectively. Kim [37] developed a novel method to design an optimized header configuration to minimize flow maldistribution using an explicit optimization algorithm and three-dimensional CFD. The optimized model improved the surface heat transfer coefficient by 7.81% compared to the base model due to the reduction in flow maldistribution. Kumar and Singh [38] numerically studied mini-channel heat sinks with modified inlet flow angles of 90°, 105°, and 120° in the inlet header. They found that the heat sink with the inlet flow angle of 105° provided the best thermal performance and more uniform cooling due to decreased flow maldistribution. Dabrowski [39] introduced a 0.5 mm threshold in rectangular mini-channels and minigap sections with rectangular, trapezoidal, triangular manifolds in Z-type flow configuration to reduce flow maldistribution. The introduction of the threshold reduces the maldistribution factor to about twice in the minigap section and to around three times in the mini-channel compared to the conventional design without the threshold. Song et al. [40] used staggered pin-fin arrays in the inlet header of a mini-channel heat sink to enhance the flow uniformity in channels. Two different headers, trapezoidal and rectangular headers with ten different pin-fin arrangements, were analyzed numerically. The results showed that the optimized pin-fin arrangement in both the inlet headers significantly enhanced the flow uniformity within the channels. Gilmore

et al. [41] used parametrically optimized elliptical pins to regulate channel flow distribution. The numerical results showed that the pins reduced the variation of channel flow rates from 19 to 1%, which is validated by experimental results of reduction of the channel flow rate variation from 20 to 7%. Li et al. [42] developed an iterative optimization algorithm to adjust the channel inlet widths for enhanced flow distribution in a parallel mini-channel heat sink with non-uniform multiple-peak heat flux.

A few attempts have been made to develop predictive tools for the quantitative assessment of flow maldistribution in designing heat sinks. Maharudrayya et al. [43] proposed an analytical model to evaluate the degree of flow distribution in parallel channels with two different inlet/outlet arrangements (Z- and U-types). The analytical model was constructed with dimensionless groups composed of friction factor, Reynolds number, and various geometric parameters (channel hydraulic diameter, header length, width, and channel depth). Numerical simulations were performed by Kim et al. [44] to investigate the influence of the distance between the inlet and outlet connection tube along the y-axis, d , header length, L_h , and header width, W_h , on flow distribution. A dimensionless geometric parameter, G , was proposed based on the simulation results for the optimal heat sink design with minimal flow maldistribution. Habib et al. [45] numerically investigated the effect of the number of inlet connection tubes, diameter of the inlet connection tube, and shape of the inlet connection tube (a circular shape and two elliptical shapes) on flow distribution. Based on their numerical simulation results, correlations for maximum mass flow rate variation were developed using the geometric parameters examined.

From the literature review, it is apparent that most of the prior studies are devoted to providing strategic frameworks for designing mini/micro-channel heat sinks by exploring the effect of heat sink design parameters on flow distribution in channels. They offer valuable insight into the relationship between several design parameters and flow maldistribution. However, the results are limited in quantitatively predicting flow maldistribution in channels. Although a few studies have been conducted to develop predictive tools for quantified assessment of flow maldistribution, valuable results are still sparse. Therefore, it is essential to quantitatively predict the flow maldistribution for design optimization of the mini-channel heat sink. Motivated by the lack of reliable tools for predicting flow maldistribution, the present study aims to offer a new flow maldistribution correlation constructed with dimensionless parameters in pursuit of quantitative prediction of flow maldistribution in mini-channel heat sinks. Five design parameters of the mini-channel heat sink are considered: channel height, channel width, header length, connection tube diameter, and total volumetric flow rate. Thirty-six different heat sinks designed with various combinations of the design parameters are numerically tested to investigate the effect of each design parameter on the uniformity of flow distribution in channels.

2. Numerical modeling

Detailed descriptions of the design parameters of the thirty-six different heat sinks used in the present study are discussed. The numerical methodology, numerical domain, and boundary and inlet conditions are reported. The grid independence test and validation of the numerical simulations are conducted to evidence the simulation result.

2.1. Model descriptions

The effect of several design parameters of mini-channel heat sinks on flow maldistribution is numerically investigated. The value of five design parameters examined in the present study, including the channel height, channel width, header length, connection tube diameter, and total volumetric flow rate in the inlet connection tube, are listed in Table 1. The channel length, L_{ch} , and header width, W_h , are constant to be 80 mm and 72 mm, respectively, for all the cases, which are typical

Table 1

Design parameters of mini-channel heat sink used for the present study.

Parameter	Description	Value
H_{ch} (mm)	Channel height	2, 4, 6
W_{ch} (mm)	Channel width	1, 2, 3
L_h (mm)	Header length	20, 30, 40
D_{in} (mm)	Connection tube diameter	6, 9, 12
V_{in} (LPM)	Total volumetric flow rate	2, 4, 6
L_{ch} (mm)	Channel length	80
W_h (mm)	Header width	72

values in the mini-channel heat sinks used to cool electronic equipment. Moreover, to comprehend the flow distribution effect according to the change in channel width, W_{ch} , the W_h is kept constant. The mini-channel heat sinks are designed with a rectangular header, a commonly used heat sink header design in various cooling applications [23,27,46,47]. Furthermore, the I-type inlet/outlet arrangement is used, which is known to provide a relatively uniform flow distribution [23–25,27]. The three-dimensional fluid domain consists of multiple parallel mini-channels, inlet/outlet headers, and inlet/outlet connection tubes. The inlet and outlet header have identical dimensions, and the central axis of connection tubes is located at the center of headers in the xy-plane (see Fig. 3(a)). For the convenience of description, the channel numbers are indexed from 1 to N along the y-direction, as illustrated in Fig. 3(b). Also shown in Fig. 3(b) is that the parallel mini-channels are equidistantly spaced from each adjacent channel with a spacing of fin width, W_f . Header width, W_h , can be related to the number of channels,

N , according to the following relation

$$W_h = NW_{ch} + (N - 1)W_f + 2W_{f,e}. \quad (1)$$

where the channel width, W_{ch} , has the same dimension as the fin width, W_f . The end wall width, $W_{f,e}$, is assumed to be half of the fin width, W_f (i. e., $W_{f,e} = 0.5W_f$). Table 2 shows the changes in the design parameter values of the mini-channel heat sink considered in the present study, which forms thirty-six cases. Table 2 also consists of the computed Reynolds number averaged over entire channels, maldistribution factor, and total pressure drop of these thirty-six cases.

2.2. Numerical method

The numerical computation has been performed with the finite volume based commercial CFD software package, Ansys Fluent 18.0 [48]. The total volumetric flow rate in the inlet connection tube, V_{in} , is calculated by assuming a uniform velocity profile at the inlet

$$V_{in} = \frac{\pi D_{in}^2 U_{in}}{4} \quad (2)$$

and the Reynolds number at the inlet connection tube, Re_{in} is defined as

$$Re_{in} = \frac{\rho U_{in} D_{in}}{\mu}. \quad (3)$$

Re_{in} calculated according to Eqs. (2) and (3) is in the range of 3527–21,161, which are much larger than 2000, indicating a turbulent

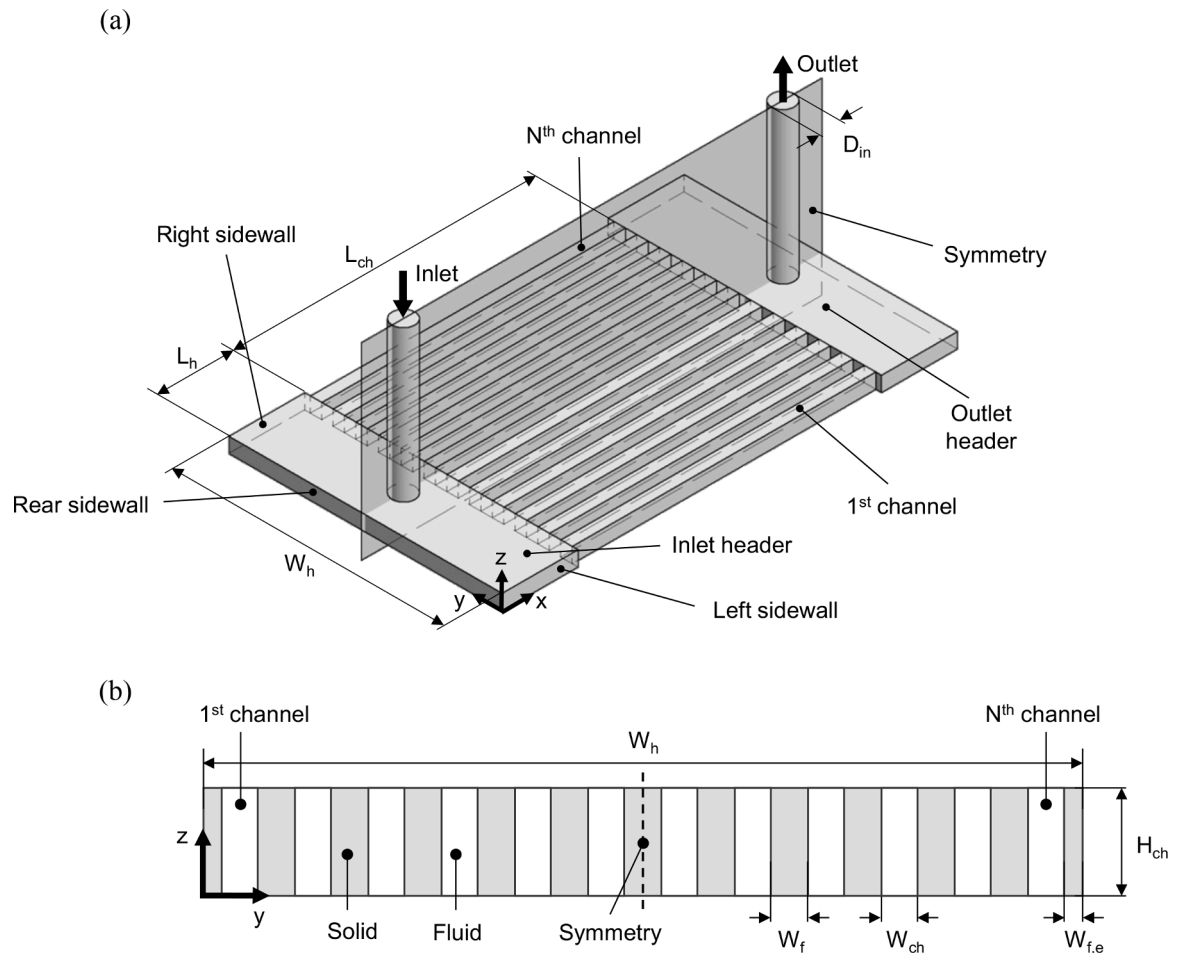


Fig. 3. Schematic of numerical domain. (a) Isometric view of three-dimensional I-type mini-channel heat sink with rectangular header. (b) Cross-sectional view of parallel mini-channels. Descriptions of the heat sink parameters are illustrated. The symmetry of the domain is utilized to model only half of the geometry for the simulation.

Table 2

Geometric configurations and computed Reynolds number averaged over entire channels, maldistribution factor, and total pressure drop for thirty-six different heat sink designs.

Case	H_{ch} (mm)	W_{ch} (mm)	D_f (mm)	L_h (mm)	D_{in} (mm)	V_{in} (LPM)	N (-)	$Re_{ch,avg}$ (-)	MF (%)	ΔP_{tot} (kPa)
1	2	1	1.3	20	6	6	36	1847	7.2	19.8
2	2	1	1.3	40	6	6	36	1847	0.8	18.6
3	2	1	1.3	20	9	6	36	1847	5.7	9.8
4	2	1	1.3	40	9	6	36	1847	0.7	9.3
5	2	1	1.3	20	12	6	36	1847	4.8	7.4
6	2	1	1.3	40	12	6	36	1847	0.5	7.0
7	2	2	2.0	30	9	4	18	1847	2.4	3.5
8	2	2	2.0	20	6	6	18	2770	9.6	17.9
9	4	1	1.6	20	6	6	36	1108	18.1	11.0
10	4	1	1.6	40	6	6	36	1108	2.5	10.2
11	4	2	2.7	40	6	2	18	616	1.7	1.1
12	4	2	2.7	20	9	2	18	616	9.0	0.4
13	4	2	2.7	40	9	2	18	616	1.2	0.4
14	4	2	2.7	20	12	2	18	616	6.4	0.2
15	4	2	2.7	40	12	2	18	616	1.2	0.2
16	4	2	2.7	40	6	4	18	1231	2.8	4.3
17	4	2	2.7	20	9	4	18	1231	11.4	1.4
18	4	2	2.7	30	9	4	18	1231	4.4	1.3
19	4	2	2.7	40	9	4	18	1231	2.1	1.3
20	4	2	2.7	20	12	4	18	1231	8.5	0.8
21	4	2	2.7	30	12	4	18	1231	3.1	0.8
22	4	2	2.7	40	12	4	18	1231	1.7	0.7
23	4	2	2.7	20	6	6	18	1847	22.4	10.0
24	4	2	2.7	30	6	6	18	1847	10.4	9.7
25	4	2	2.7	40	6	6	18	1847	3.9	9.6
26	4	2	2.7	20	9	6	18	1847	13.8	3.1
27	4	2	2.7	40	9	6	18	1847	2.4	2.9
28	4	2	2.7	20	12	6	18	1847	9.8	1.7
29	4	2	2.7	40	12	6	18	1847	2.2	1.2
30	4	3	3.4	20	6	6	12	2374	22.6	9.9
31	4	3	3.4	40	6	6	12	2374	4.1	9.5
32	6	1	1.7	20	6	6	36	791	22.8	9.2
33	6	2	3.0	30	9	4	18	923	10.7	1.0
34	6	2	3.0	20	6	6	18	1385	32.9	8.8
35	6	3	4.0	20	6	6	12	1847	33.0	8.7
36	6	3	4.0	40	12	6	12	1847	5.1	0.9

flow regime. Table 2 shows the values of channel Reynolds number averaged over entire mini-channels, $Re_{ch,avg}$, are in the range of 616–2770. Although most of $Re_{ch,avg}$ correspond to the laminar or transition flow regime, a turbulent flow regime can be encountered in some channels due to flow maldistribution. Moreover, local turbulent regions can exist in the inlet/outlet connection tubes and headers. For this reason, the turbulent flow condition is considered by using a realizable k - ϵ turbulence model with enhanced wall treatment [48,49]. The Semi-Implicit Method for Pressure-Linked Equations-Consistent (SIM-PLEC) algorithm [50] is used to solve the pressure-velocity coupling. The second-order and second-order upwind schemes are used for the spatial discretization of pressure and momentum, respectively. The governing equations used in the numerical simulation are the Reynolds Averaged Navier Stokes (RANS) equations for steady-state and incompressible flow. The time-averaged continuity and momentum equations can be written in cartesian coordinates as continuity equation [48]:

$$\frac{\partial U_j}{\partial x_j} = 0 \quad (4)$$

momentum equation [48]:

$$\rho \frac{\partial}{\partial x_j} (U_i U_j) = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} \left((\mu + \mu_t) \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) \right) \quad (5)$$

and transport equations for k and ϵ can be expressed as turbulent kinetic energy equation [51]:

$$\rho \frac{\partial}{\partial x_i} (k U_i) = \frac{\partial}{\partial x_j} \left(\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right) + G_k - \rho \epsilon \quad (6)$$

turbulent dissipation rate equation [51]:

$$\rho \frac{\partial}{\partial x_i} (\epsilon U_i) = \frac{\partial}{\partial x_j} \left(\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right) + \rho C_1 S \epsilon - \rho C_2 \frac{\epsilon^2}{k + \sqrt{\nu \epsilon}} \quad (7)$$

The constants applied in Eqs. (6) and (7) are $\sigma_k = 1.0$, $\sigma_\epsilon = 1.2$, and $C_2 = 1.9$, and C_1 is given by

$$C_1 = \max \left[0.43, \frac{\eta}{\eta + 5} \right] \quad (8)$$

where

$$\eta = \frac{S k}{\epsilon}, \quad S = \sqrt{2 S_{ij} S_{ij}} \quad (9)$$

To utilize the geometric symmetry of heat sinks, only half of the domain is modeled by imposing a symmetry boundary condition at the symmetry plane, as illustrated in Fig. 3(a). No-slip boundary condition is applied to all interfaces between fluid and solid regions, and atmospheric pressure is specified at the outlet connection tube. Water with $\rho = 998.21 \text{ kg/m}^3$ and $\mu = 0.001002 \text{ kg/m-s}$ at 20°C , obtained from the REFPROP program [52], is used as the working fluid.

2.3. Grid independence test and validation

Structured grids for headers and channels and unstructured grids for connection tubes construct the computational fluid domains. The detailed sizes of structured grids for three different grid systems are summarized in Table 3. The grid independence test is conducted on case 9, which has 20 mm long headers and thirty-six parallel rectangular channels with 1 mm width and 4 mm depth. Three grid systems with 1.6×10^6 cells (grid system 1), 4.0×10^6 cells (grid system 2), and 6.7×10^6 cells (grid system 3) are used for the grid independence test.

Table 3

Grid size of three different grid systems used for grid independence test.

	d_x (m)	d_y (m)	d_z (m)
Grid system 1	4.0×10^{-4}	1.0×10^{-4}	2.0×10^{-4}
Grid system 2	3.3×10^{-4}	7.5×10^{-5}	1.6×10^{-4}
Grid system 3	2.0×10^{-4}	6.3×10^{-5}	1.3×10^{-4}

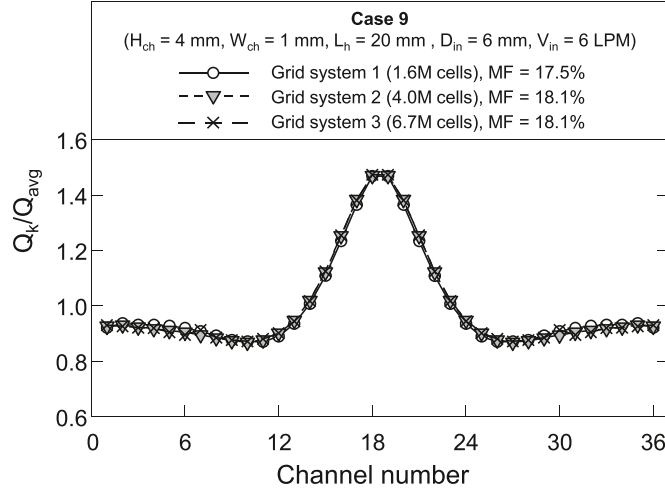


Fig. 4. Normalized mass flow rate of each channel for case 9 ($H_{ch} = 4$ mm, $W_{ch} = 1$ mm, $L_h = 20$ mm, $D_{in} = 6$ mm, $V_{in} = 6$ LPM) with different grid systems. The maldistribution factor (MF) values for the different grid systems are also given. Negligible differences in the normalized mass flow rate are observed for all three grid systems. Since the grid systems 2 and 3 have identical MF values of 18.1%, all the subsequent calculations are carried out using grid system 2 with about 4.0×10^6 cells.

Fig. 4 shows the influence of grid size on the simulation results is examined by comparing the normalized mass flow rate of each channel. Seemingly insignificant differences among the three simulation results can be observed in Fig. 4. Moreover, the maldistribution factor, MF , can be used for comparison purposes, defined as

$$MF (\%) = \sqrt{\frac{1}{N-1} \sum_{k=1}^N \left(\frac{Q_k - Q_{avg}}{Q_{avg}} \right)^2} \times 100 \quad (10)$$

where the mass flow rate averaged over the entire mini-channels, Q_{avg} , is given by

$$Q_{avg} = \sum_{k=1}^N \frac{Q_k}{N}. \quad (11)$$

Notice that the MF is expressed as the standard deviation of the normalized mass flow rates of channels. Thus, a smaller MF value indicates a more uniform flow distribution in channels. The value of MF for the three grid systems 1, 2, and 3 are 17.5%, 18.1%, and 18.1%, respectively. Since the grid systems 2 and 3 have identical MF values of 18.1%, all the subsequent calculations are carried out using grid system 2 with about 4.0×10^6 cells. Fig. 5 shows the grid structure of case 9 with grid system 2.

The validation of the numerical simulation is accomplished by comparing the computed pressure drop results to predictions from previous correlations. The following relation is used to predict the pressure drop in each channel,

$$\Delta P_{ch} = \frac{2f_{app} \rho \bar{U}^2 L_{ch}}{D_f} \quad (12)$$

The apparent friction factor spanning both the laminar developing

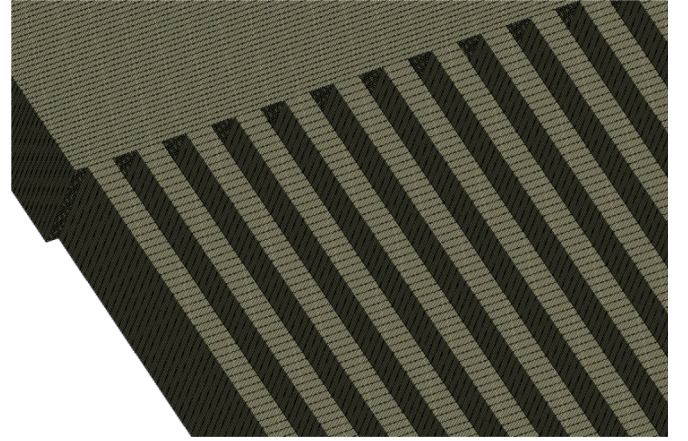


Fig. 5. Grid structure of fluid domain for case 9 with grid system 2 (4.0×10^6 cells). Fully structured grid with quadrilateral cells is created for the entire numerical domain.

and developed flow regions, f_{app} , in Eq. (12) is given by Copeland [53],

$$f_{app} = \frac{\left\{ \left[3.2 \left(\frac{L_{ch}}{Re D_f} \right)^{-0.57} \right]^2 + (f_d Re)^2 \right\}^{1/2}}{Re} \quad (13)$$

where Re refers to the Reynolds number in each channel calculated from simulation results, and f_d is the friction factor for the laminar fully-developed flow in a rectangular channel, which is offered by Shah and London [54] as,

$$f_d Re = 24(1 - 1.3553\beta + 1.9467\beta^2 - 1.7012\beta^3 + 0.9564\beta^4 - 0.2537\beta^5). \quad (14)$$

Eqs. (12)–(14) show that the pressure drop in the channel is influenced by channel hydraulic diameter, D_f , and average velocity in the channel, $\bar{U}$. Hence, validation assessments performed by comparing the pressure drop in each channel can prove more effective by considering a wider range of design parameters related to D_f and $\bar{U}$. For this reason, case 11 and case 36, which have different values of H_{ch} , W_{ch} , and V_{in} , are chosen for validation. The computed and predicted pressure drop in each channel for both case 11 and case 36 are compared in Fig. 6. The discrepancy between computed and predicted results for pressure drop in each channel can be calculated using mean absolute error (MAE), which is defined as

$$MAE (\%) = \frac{1}{N} \sum_{k=1}^N \left| \frac{\Delta P_{k,pred} - \Delta P_{k,comp}}{\Delta P_{k,comp}} \right| \times 100 \quad (15)$$

With an MAE of 5.7% and 9.3% for case 11 and case 36, respectively, reasonable agreement is demonstrated between the results computed by the three-dimensional numerical simulations and predicted by previous correlations.

3. Results and discussion

Initially, the physics of the flow distribution in the mini-channel heat sink is discussed to comprehend the flow behavior in the heat sink. Then, the effects of the heat sink parameters considered in the present study are analyzed and discussed in detail to form a profound understating of how and to what extent each parameter affects the flow uniformity and causes flow maldistribution in the channels of the mini-channel heat sink. Through this knowledge, a new correlation for the maldistribution factor is formulated with dimensionless groups of different parameters to quantitatively assess the flow maldistribution in channels for designing mini-channel heat sinks. Finally, the parametric assessment of the effects of the different parameters on the flow maldistribution using

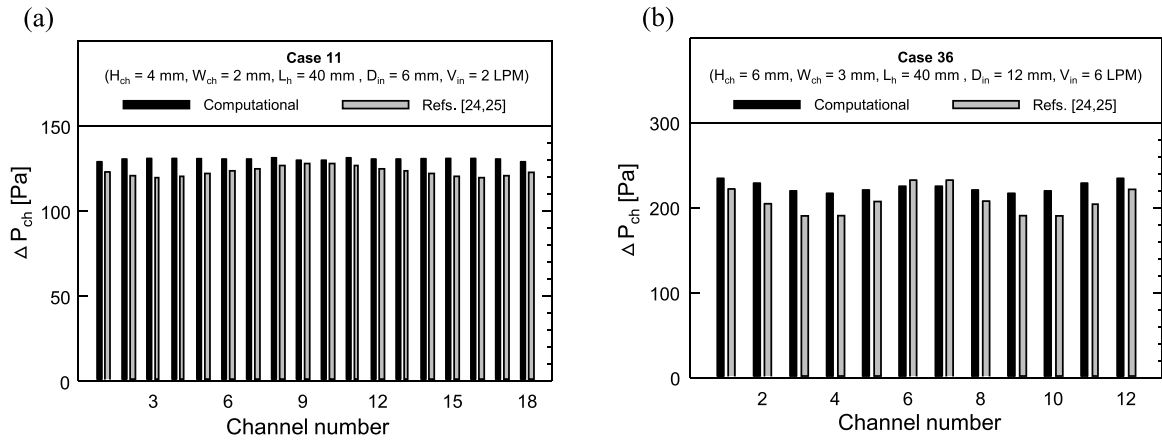


Fig. 6. Comparison of computed and predicted pressure drop in each channel of heat sink for (a) case 11 and (b) case 36. The discrepancy between computed and predicted results for pressure drop in each channel is calculated using mean absolute error (MAE). An MAE of 5.7% and 9.3% for case 11 and case 36, respectively, demonstrate reasonable agreement between the simulated and predicted pressure drop.

the new correlation is performed.

3.1. Flow distribution characteristics

Fig. 7 shows the flow distribution characteristics in the inlet header and channels for case 9, which is the typical flow behavior observed in all the cases. The streamlines represent the overall flow pattern. The incoming jet flow from the inlet connection tube impinges on the inlet header, and the fluid is distributed to multiple parallel channels. The fluid spreads out in every direction of the inlet header after impacting the xy-plane in the inlet header, and each flow streamlines directly follow the shortest flow path to the channel entrance, as shown by the solid arrows in Fig. 7(a). Hence, the distance from the stagnation region to the channel entrance significantly affects the channel flow distribution. This explains the largest flow rate in the 18th channel near the symmetry plane, whose entrance is the closest to the inlet connection tube, as shown in Fig. 4. The rear sidewall then confines the flow, and the fluid begins to circulate along the rear sidewall. The flow is separated from the wall by the corner vortex near the rear sidewall and the left sidewall. As a result, the flow stream primarily enters the channels 1–4 along the left sidewall, as shown by the solid arrow in Fig. 7(b). This flow behavior is responsible for the fluid supply to the channels far from the inlet connection tube, where only a small amount of fluid is supplied. In most cases, the channels near the inlet connection tube receive most of the bulk flow because of the comparatively high flow velocity near the

stagnation region due to the flow impingement effect. As a result, the channels far from the inlet connection tube receive less flow, resulting in flow maldistribution. Hence, to minimize flow maldistribution attributed to the flow features mentioned above, it is paramount to identify the critical design parameters of the mini-channel heat sink that affect the flow behaviors.

The individual pressure drop in channels and headers and the total pressure drop across the entire heat sink greatly influence flow distribution. In general, the pressure drop distribution among parallel channels becomes more uniform when the ratio of the averaged channel pressure drop to the total pressure drop, $\Delta P_{ch,avg}/\Delta P_{tot}$, is large enough, resulting in a more uniform flow distribution, as discussed by [55]. Therefore, it is also vital to examine the effect of different design parameters on pressure drop components.

3.2. Effects of channel height and width

The variations of MF for different H_{ch} and W_{ch} are presented in Fig. 8 (a) and (b), respectively. The figures show a clear trend of increasing MF with the increase in H_{ch} and W_{ch} . These results agree with previous studies [55–57], which reported that a smaller channel hydraulic diameter offers a more uniform flow distribution in channels. Fig. 9(a) and (b) provide detailed descriptions of the relationship between the channel average pressure drop, total pressure drop, $\Delta P_{ch,avg}/\Delta P_{tot}$, and MF for different H_{ch} and W_{ch} , respectively. Both figures demonstrate that

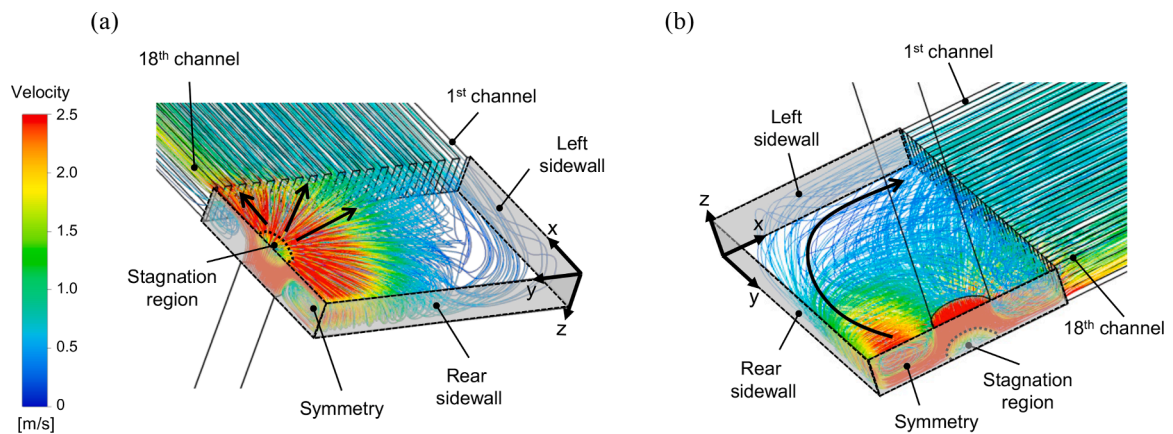


Fig. 7. Streamlines in inlet header of case 9 ($H_{ch} = 4$ mm, $W_{ch} = 1$ mm, $L_h = 20$ mm, $D_{in} = 6$ mm, $V_{in} = 6$ LPM), (a) bottom view and (b) top view. The solid arrows in the header represent the flow distribution behavior. The channels close to the inlet connection tube receive most of the bulk flow. The distance from the stagnation region to the channel entrance significantly affects the channel flow distribution resulting in flow maldistribution.

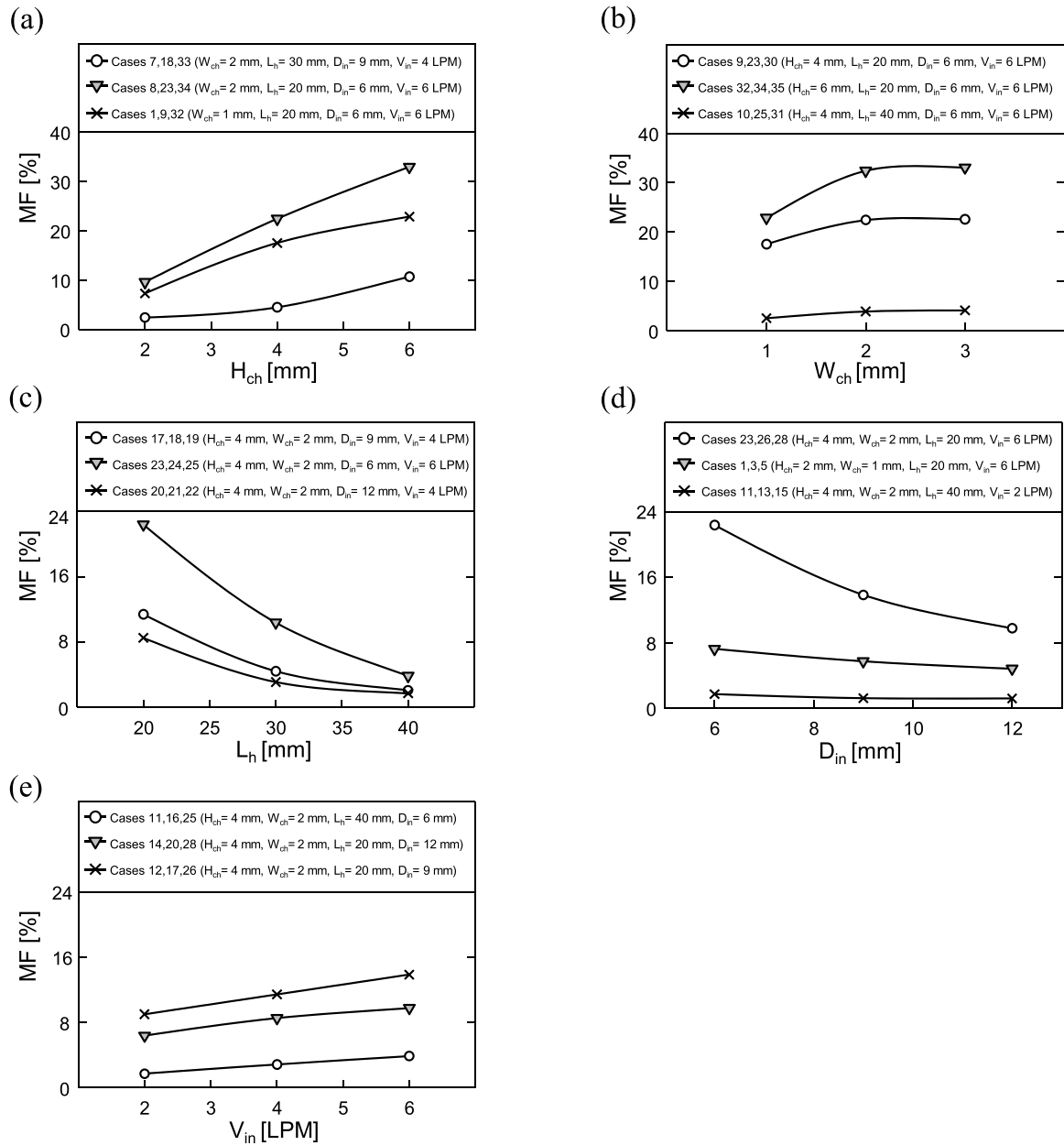


Fig. 8. Variations of maldistribution factor for different values of: (a) H_{ch} , (b) W_{ch} , (c) L_h , (d) D_{in} and (e) V_{in} . The MF increases with an increase in H_{ch} , W_{ch} , and V_{in} and decreases with an increase in L_h and D_{in} .

an increase in the channel hydraulic diameter decreases both the channel average pressure drop and total pressure drop leading to the decrease in $\Delta P_{ch,avg}/\Delta P_{tot}$, resulting in a severe flow maldistribution in the channels. Notice that because the header height is the same as that of the channel, the change of H_{ch} affects not only pressure drop in channels but also in headers. Nonetheless, Fig. 9(a) shows that the rate of increase in MF per H_{ch} rises significantly as H_{ch} increases due to the continuous decline of $\Delta P_{ch,avg}/\Delta P_{tot}$ with increasing H_{ch} . In Fig. 9(b), as W_{ch} increases, $\Delta P_{ch,avg}/\Delta P_{tot}$ decreases rapidly up to $W_{ch} = 2$ mm, and $\Delta P_{ch,avg}/\Delta P_{tot}$ becomes nearly constant at $2 \text{ mm} < W_{ch} < 3 \text{ mm}$. Hence, the rate of increase in MF per W_{ch} is significantly reduced at higher W_{ch} due to the $\Delta P_{ch,avg}/\Delta P_{tot}$ becoming nearly constant at higher W_{ch} . The increase in H_{ch} and W_{ch} increases the cross-sectional flow area causing the decrease in $\Delta P_{ch,avg}/\Delta P_{tot}$. However, for a fixed header width, Eq. (1) shows that the number of channels, N , decreases as W_{ch} increases, thereby increasing the flow rate of each channel. Hence, the increased flow rate in the channels increases the pressure drop, which counteracts

the decrease in $\Delta P_{ch,avg}/\Delta P_{tot}$ caused by the increased cross-sectional area leading to the nearly constant value of $\Delta P_{ch,avg}/\Delta P_{tot}$, as well as MF at higher W_{ch} .

Overall, decreasing H_{ch} and W_{ch} provides more uniform flow distribution in channels with an increase in total pressure drop. A higher total pressure drop consequently increases the required pumping power of the system. Hence, determining a suitable value of H_{ch} and W_{ch} is essential to enhance flow uniformity while avoiding a significant increase in total pressure drop.

3.3. Effect of header length

Fig. 8(c) shows the general trend that MF decreases with an increase in L_h . To comprehensively examine the reason for the effect of L_h on flow maldistribution, the normalized mass flow rate of each channel for three different cases (i.e., cases 23, 24, and 25) are compared in Fig. 10. The channels 10–18 are symmetry of the channels 1–9. The channels 9 and

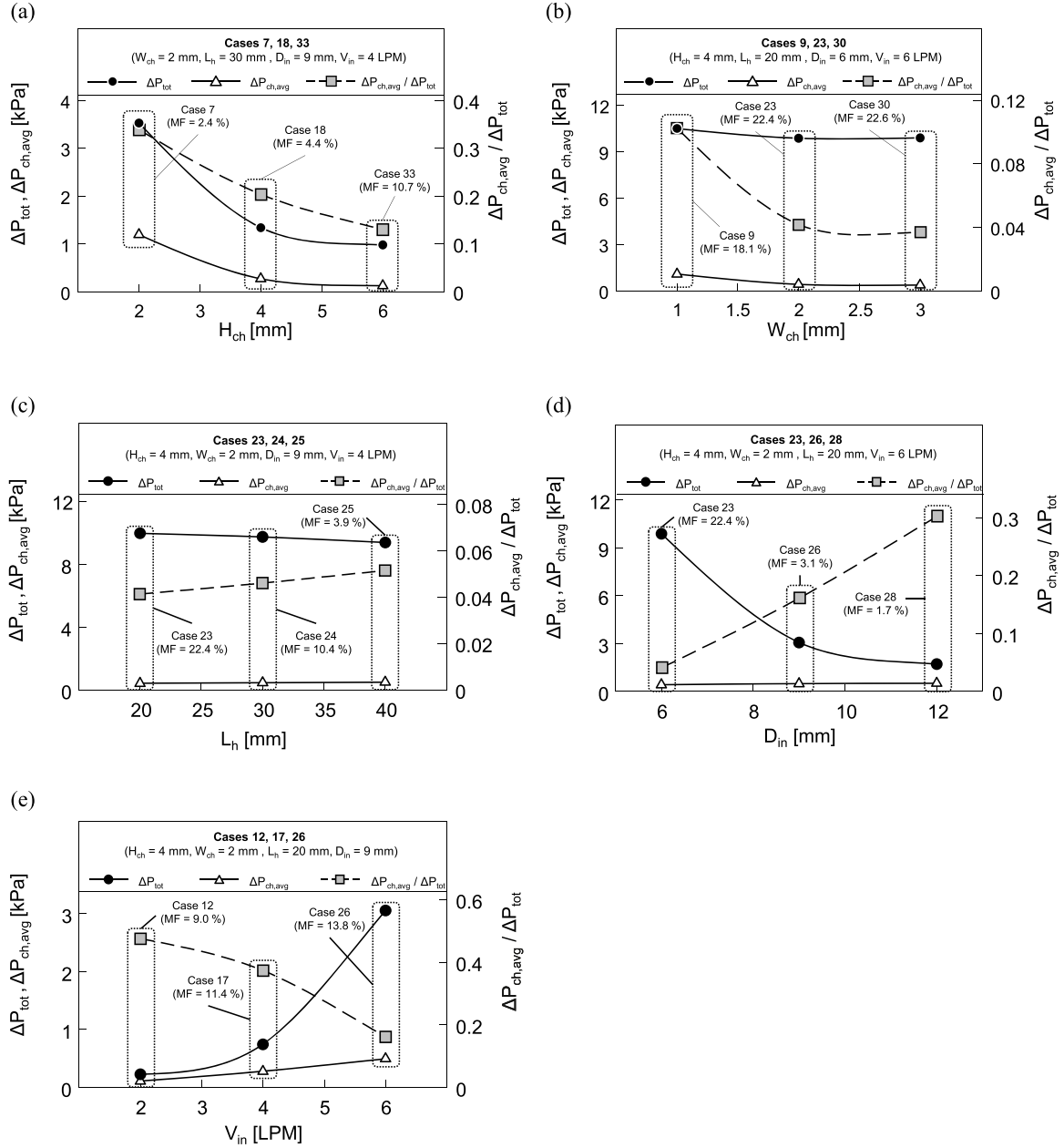


Fig. 9. Variations of total pressure drop and channel average pressure drop, and relationship between the ratio of averaged channel pressure drop to total pressure drop and flow maldistribution for different values of (a) H_{ch} , (b) W_{ch} , (c) L_h , (d) D_{in} and (e) V_{in} . A clear trend that MF is inversely proportional to $\Delta P_{ch,avg} / \Delta P_{tot}$ is observed.

10 are the center channels. Fig. 10 shows that the maximum peak flow rates occur at the center channels for all three L_h values due to the proximity of these channels to the inlet connection tube. As the L_h increases, the peak flow rates in and near the center channels decrease and the amount of fluid distributed to the rest of the channels increases, resulting in more uniform flow distribution. This can be explained by comparing the flow streamlines in the inlet header and channels for the three cases in Fig. 11(a). High flow velocity appears in the center channels for all three cases since the entrances of these channels are close to the inlet connection tube. However, as L_h increases, the flow velocity becomes low at the center channel entrance due to the extended header length. Therefore, relatively more fluid is distributed to the rest of the channels. Fig. 11(b) presents the velocity vectors of three cases in the region near the parallel channels taken at $z = 1 \text{ mm}$. In case 23, with the lowest L_h value, the bulk of the high-velocity flow enters the channels 7–9 since these channels are closer to the stagnation region. As the

L_h increases, the distance from the stagnation region to the channels increases, and the bulk fluid flow circulates in the inlet header before entering the channels resulting in a more uniformly distributed flow.

Fig. 9(c) shows that increasing L_h causes a decrease in ΔP_{tot} and a nearly constant $\Delta P_{ch,avg}$, thereby increasing $\Delta P_{ch,avg} / \Delta P_{tot}$. However, although the figure shows the general trend of decreasing MF with increasing $\Delta P_{ch,avg} / \Delta P_{tot}$, a strong relationship between $\Delta P_{ch,avg} / \Delta P_{tot}$ and MF is not found for varying L_h , given that only a slight increase of $\Delta P_{ch,avg} / \Delta P_{tot}$ is observed while the difference of MF values among the cases 23, 24, and 25 is more than 80%. Overall, L_h is a crucial design parameter that enhances flow uniformity in channels without a significant increase in total pressure drop, as shown in Fig. 9(c). Nevertheless, heat sinks with a larger header length often encounter compactness issues. Hence, determining a suitable L_h value considering the best compromise between flow uniformity in channels and compactness of the mini-channel heat sink is essential.

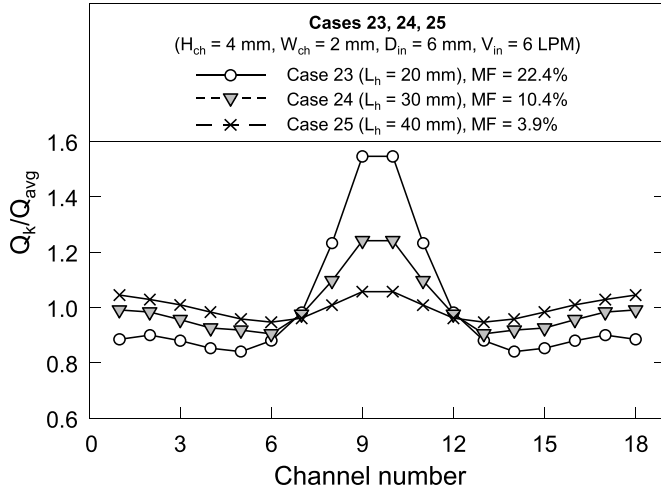


Fig. 10. Normalized mass flow rate of each channel for heat sinks having different header lengths. As the L_h increases, the peak flow rates in and near the center channels decrease and the amount of fluid distributed to the rest of the channels increases, resulting in more uniform flow distribution.

3.4. Effect of connection tube diameter

Fig. 8(d) shows the general trend that MF decreases as D_{in} increases. Fig. 9(d) presents the variations of $\Delta P_{ch,avg}/\Delta P_{tot}$ and MF for three cases (i.e., cases 23, 26, and 28). The result indicates that a larger value of D_{in} contributes to a significant increase in $\Delta P_{ch,avg}/\Delta P_{tot}$, thereby leading to a more uniform flow distribution in channels. Also shown in Fig. 9(d) is that ΔP_{tot} appreciably decreases while $\Delta P_{ch,avg}$ is nearly constant with increasing D_{in} , implying that a larger value of D_{in} reduces the pressure

drop in the inlet and the outlet headers. The increase in D_{in} reduces the velocity of flow impingement, causing the reduced pressure drop in the inlet and the outlet headers. Therefore, D_{in} is a crucial design parameter that can enhance flow uniformity in channels while reducing total pressure drop.

3.5. Effect of inlet flow rate

The variations of MF for different V_{in} are presented in Fig. 8(e). The figure shows that MF increases as V_{in} increases, which agrees with previous studies [58–60]. Fig. 9(e) provides more detailed descriptions of the relationship between $\Delta P_{ch,avg}/\Delta P_{tot}$ and MF for different V_{in} . The figure shows that a larger value of V_{in} produces an increase in both $\Delta P_{ch,avg}$ and ΔP_{tot} , $\Delta P_{ch,avg}/\Delta P_{tot}$ decreases with increasing V_{in} and thereby leading to an increase in MF . This implies that the variations of V_{in} significantly influence pressure drop in headers more than in channels, given that the total pressure drop is the sum of channel pressure drop and pressure drop in inlet and outlet headers. Overall, a smaller V_{in} value reduces total pressure drop and enhances flow uniformity in channels. However, obtaining a larger heat transfer coefficient in channels can generally benefit from increasing the inlet flow rate. Hence, the value of V_{in} needs to be determined for the best compromise between flow uniformity in channels and thermal performance in the mini-channel heat sink.

3.6. Correlation for maldistribution factor

Based on the parametric analysis of the relationship between the design parameters of the heat sink and flow maldistribution in channels, a new correlation for the maldistribution factor is developed for quantitative assessment of flow maldistribution in channels for designing mini-channel heat sinks. The following expression was formulated with

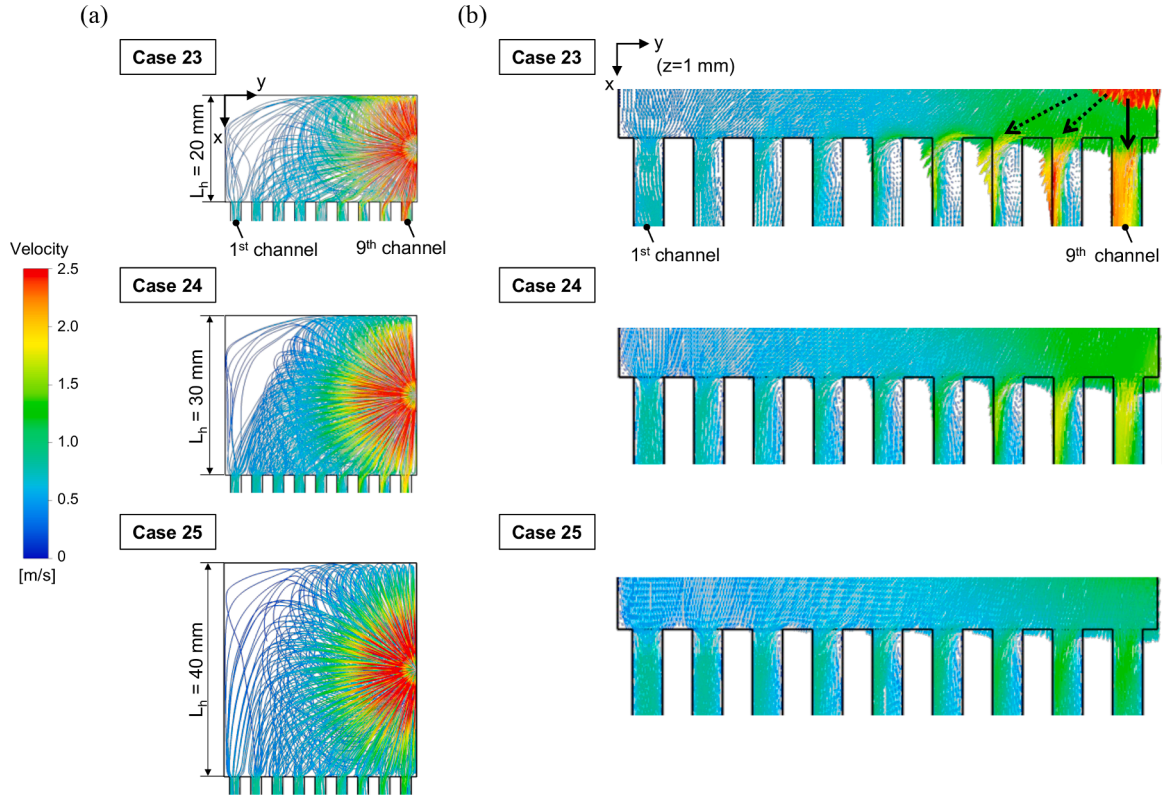


Fig. 11. (a) Streamlines in different inlet headers and (b) velocity vectors near channel entrance for different header lengths at $H_{ch} = 4$ mm, $W_{ch} = 2$ mm, $D_{in} = 6$ mm, and $V_{in} = 6$ LPM. As the L_h increases, the distance from the stagnation region to the channels increases, and the bulk fluid flow circulates in the inlet header before entering the channels, resulting in a more uniformly distributed flow.

dimensionless groups, Reynolds number in inlet connection tube, average apparent friction factor, Reynolds number averaged over entire channels, and two dimensionless geometric parameters (i.e., D_f/D_{in} and D_f/L_h). The dimensionless groups are formulated based on the numerical analysis results of the main parameters affecting MF.

$$MF (\%) = c_1 (Re_{in})^{c_2} (f_{app,avg})^{c_3} (Re_{ch,avg})^{c_4} \left(\frac{D_f}{D_{in}}\right)^{c_5} \left(\frac{D_f}{L_h}\right)^{c_6} \times \left\{ c_7 \left(\frac{D_f}{D_{in}}\right)^{c_8} + c_9 \left(\frac{D_f}{L_h}\right)^{c_{10}} \right\}^{c_{11}} \quad (16)$$

The Reynolds number averaged over entire channels, $Re_{ch,avg}$, can be calculated from the following relation

$$Re_{ch,avg} = \frac{\pi \rho U_{in} D_{in}^2 D_f}{2 \mu W_h H_{ch}} \quad (17)$$

and the average apparent friction factor, $f_{app,avg}$, can be obtained by replacing the channel Reynolds number, Re , in Eqs. (13) and (14) with

$Re_{ch,avg}$. The constants, c_1 – c_{11} , in Eq. (16) are optimized to yield least MAE versus the total of 36 numerically calculated MF values. The following correlation for maldistribution factor is then proposed,

$$MF (\%) = 2.578 (Re_{in})^{1.617} (f_{app,avg})^{-0.388} (Re_{ch,avg})^{-1.261} \left(\frac{D_f}{D_{in}}\right)^{0.547} \left(\frac{D_f}{L_h}\right)^{3.673} \times \left\{ 0.6 \left(\frac{D_f}{D_{in}}\right)^{0.55} + 6.5 \left(\frac{D_f}{L_h}\right)^{1.85} \right\}^{-3.048} \quad (18)$$

The parameter ranges used in developing the above correlation correspond to $3527 < Re_{in} < 21,162$, $616 < Re_{ch,avg} < 2770$, $0.112 < D_f/D_{in} < 0.667$, and $0.033 < D_f/L_h \leq 0.2$. It is found that the correlation deviates from the entire numerical data by a root-mean-squared-error (RMSE) of 1.0%, which is defined as

$$RMSE (\%) = \sqrt{\frac{1}{n} \sum (MF_{comp} - MF_{pred})^2} \quad (19)$$

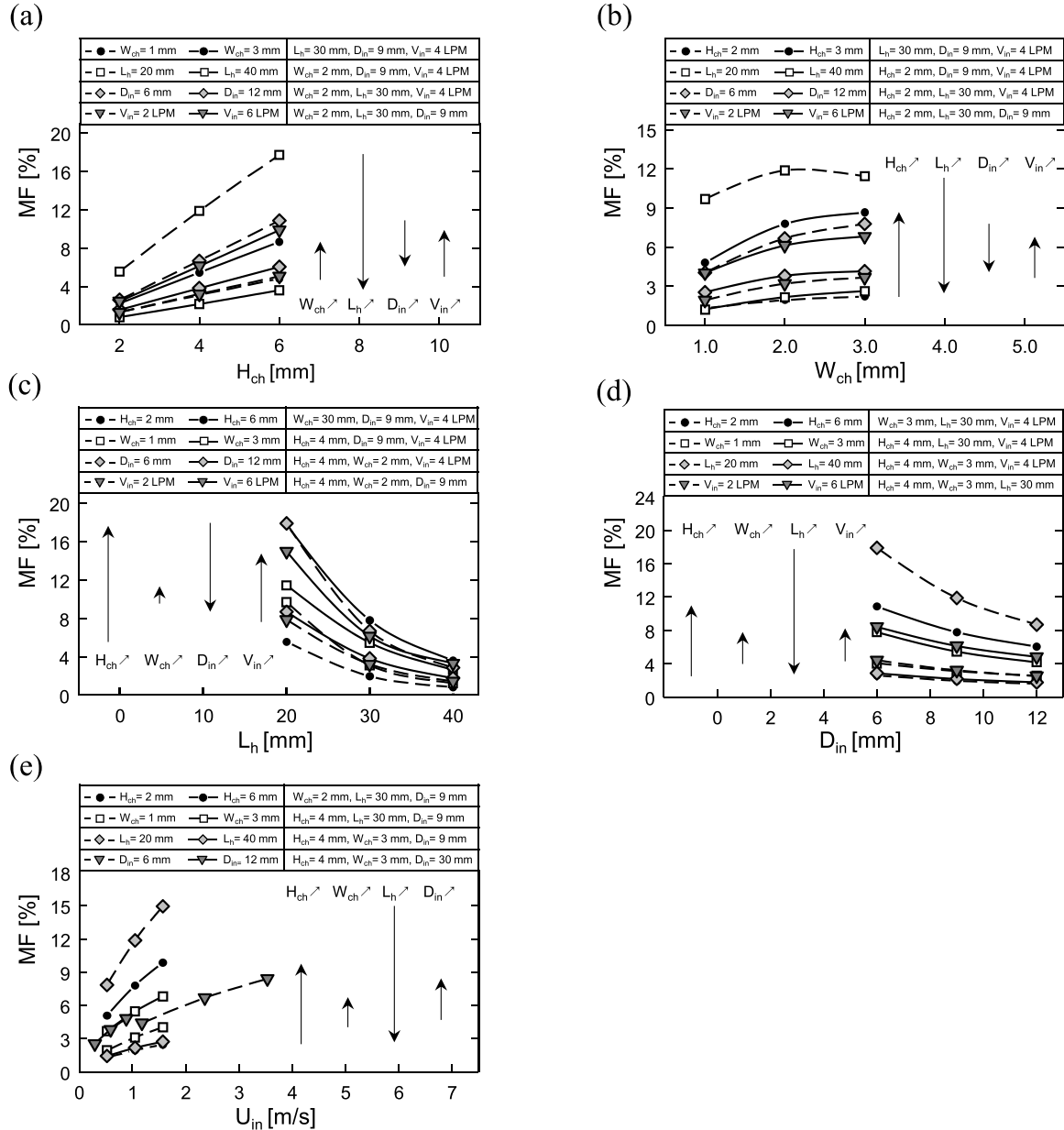


Fig. 12. Parametric trend presenting relative influence of design parameters on flow maldistribution using new correlation: (a) H_{ch} , (b) W_{ch} , (c) L_h , (d) D_{in} and (e) U_{in} . The magnitude of the relative influence of each variable on flow maldistribution can be expressed in the following order: $L_h > H_{ch} > D_{in} > V_{in} > W_{ch}$.

Fig. 12 shows a parametric assessment of the effects of H_{ch} , W_{ch} , L_h , D_{in} , and U_{in} on flow maldistribution using the new correlation. Overall, MF increases with increasing H_{ch} , W_{ch} , and U_{in} , as shown in Fig. 12(a), (b), and (e), respectively, whereas MF decreases with increasing L_h and D_{in} as shown in Fig. 12(c) and (d), respectively. These results agree with the parametric trend captured from numerical simulations. Notice in Fig. 12(b) the slight change in the trend of MF versus W_{ch} for $L_h = 20$ mm: MF increases with increasing W_{ch} at $1\text{ mm} < W_{ch} < 2\text{ mm}$ then decreases at $2\text{ mm} < W_{ch} < 3\text{ mm}$. This non-monotonic variation of MF versus W_{ch} is similar to the simulation results discussed in Section 3.2 (see Fig. 8(b)).

The maximum differences of MF for each design parameter are indicated with arrows in Fig. 12 to compare the relative influence of each design parameter on flow maldistribution. In Fig. 12(a), (b), (d), and (e), the magnitude of the arrow is the largest for L_h , indicating that L_h has the most substantial influence on flow maldistribution in the parameter range. On the other hand, the influence of W_{ch} on flow maldistribution is not more significant than that of other design parameters, as indicated by the smallest magnitude of the arrows in Fig. 12(a), (c), and (e). In this manner, the magnitude of the relative influence of each variable on flow maldistribution can be expressed in the following order: $L_h > H_{ch} > D_{in} > V_{in} > W_{ch}$.

3.7. Limitations and future work

The present study numerically investigates the influence of design parameters of mini-channel heat sinks on flow distribution to develop a correlation for predicting the maldistribution factor. It is crucial to address the potential limitations of the present study, which could be addressed in future research. Validation with experimental data is critical despite validating the present numerical model with previous correlations. Moreover, validating the developed maldistribution factor correlation with other corresponding mini-channel heat sink data is essential. The present study considered the design parameters independent. Hence, the investigation of the ratio of the design parameters is also necessary. The present study is specific to heat sinks with an I-type (vertical) inlet/outlet arrangement located at the center of a rectangular inlet/outlet header within a particular base area range used in cooling specific electronic equipment such as insulated-gate bipolar transistor (IGBT) power modules, and computer CPUs. Hence, it is essential to develop a universal correlation to predict the maldistribution factor in mini-channel heat sinks by considering the limitations of the present study.

4. Conclusions

The present study explored the influence of design parameters of mini-channel heat sinks on flow distribution to develop a new correlation for predicting the maldistribution factor. Five design parameters, channel height, channel width, header length, connection tube diameter, and total volumetric flow rate, were considered, and thirty-six different heat sinks were designed to investigate flow distribution in channels numerically. The present numerical model is successfully validated by comparing the computed pressure drop with the previous correlation in the literature.

The heat sinks with I-type inlet/outlet flow arrangement exhibit a general feature; the most significant amount of fluid is distributed to the channel whose entrance is located closest to the inlet connection tube. In addition, the inlet headers observe distinctive flow behaviors attributed to the flow impingement effect and corner vortex. Primarily, the channels near the inlet connection tube receive most of the bulk flow because of the comparatively high flow velocity near the stagnation region due to the flow impingement effect. Hence, the channels far from the inlet connection tube receive less flow resulting in flow maldistribution.

In general, $\Delta P_{ch,avg}/\Delta P_{tot}$ greatly influence flow distribution in mini-channel heat sinks, and the parametric effects of design parameters on

flow maldistribution are discussed using the relationship between $\Delta P_{ch,avg}/\Delta P_{tot}$ and MF . A clear trend that MF is inversely proportional to $\Delta P_{ch,avg}/\Delta P_{tot}$ is observed for all design parameters considered in the present study. A decrease in H_{ch} , W_{ch} , and V_{in} and an increase in L_h and D_{in} contribute to increasing $\Delta P_{ch,avg}/\Delta P_{tot}$, reducing MF and enhancing channel flow uniformity. Therefore, in the design process of mini-channel heat sinks, optimizing the values of H_{ch} , W_{ch} , V_{in} , L_h , and D_{in} can provide optimum flow uniformity levels with an acceptable compromise of total pressure drop.

A new correlation for the maldistribution factor is proposed based on the numerical results to provide a predictive tool for the quantitative assessment of flow maldistribution in the design process of mini-channel heat sinks. The overall $RSME$ of this correlation of 1.0% demonstrates its excellent predictive capability for various heat sink design parameters. The relative influence of each design parameter on flow maldistribution is assessed by analyzing the parametric trend of the predicted maldistribution factor with the new correlation. The result shows that L_h and H_{ch} have the dominant influence on flow distribution in channels, while the effect of W_{ch} on flow maldistribution is relatively insignificant. The magnitude of the relative influence of each variable on flow maldistribution can be expressed in the following order: $L_h > H_{ch} > D_{in} > V_{in} > W_{ch}$.

CRediT authorship contribution statement

Jung-Youn Song: Investigation, Validation, Methodology, Software, Writing – original draft. **Santhosh Senguttuvan:** Investigation, Validation, Writing – review & editing. **Won-Woo Choi:** Validation, Software, Writing – review & editing. **Sung-Min Kim:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

No.

Data Availability

Data will be made available on request.

References

- [1] Mudawar I. Assessment of high-heat-flux thermal management schemes. *IEEE Trans Compon Packag Technol* 2001;24:122–41.
- [2] Chein R, Huang G. Analysis of microchannel heat sink performance using nanofluids. *Appl Therm Eng* 2005;25:3104–14.
- [3] Revellin R, Thome JR. A new type of diabatic flow pattern map for boiling heat transfer in microchannels. *J Micromech Microeng* 2007;17:788–96.
- [4] Wojtan L, Ursenbacher T, Thome JR. Investigation of flow boiling in horizontal tubes: part1 – a new diabatic two-phase flow pattern map. *Int J Heat Mass Transf* 2005;48:2955–69.
- [5] Shin HC, Kim SM. Generalized flow regime map for two-phase mini /micro-channel flows. *Int J Heat Mass Transf* 2022;196:123298.
- [6] Zhang H, Mudawar I, Hasan MM. CHF model for subcooled flow boiling in Earth gravity and microgravity. *Int J Heat Mass Transf* 2007;50:4039–51.
- [7] Lee H, Mudawar I, Hasan MM. Experimental and theoretical investigation of annular flow condensation in microgravity. *Int J Heat Mass Transf* 2013;61:293–309.
- [8] Balasubramanian K, Jagirdar M, Lee PS, Teo CJ, Chou SK. Experimental investigation of flow boiling heat transfer and instabilities in straight microchannels. *Int J Heat Mass Transf* 2013;66:655–71.
- [9] Lee S, Devahdhanush VS, Mudawar I. Frequency analysis of pressure oscillations in large length-to-diameter two-phase micro-channel heat sinks. *Int J Heat Mass Transf* 2018;116:273–91.
- [10] Kim SM, Mudawar I. Theoretical model for local heat transfer coefficient for annular flow boiling in circular mini/micro-channels. *Int J Heat Mass Transf* 2014;73:731–42.
- [11] Kim SM, Mudawar I. Theoretical model for annular flow condensation in rectangular micro-channels. *Int J Heat Mass Transf* 2012;55:958–70.
- [12] Kim SM, Mudawar I. Universal approach to predicting two-phase frictional pressure drop for mini/micro-channel saturated flow boiling. *Int J Heat Mass Transf* 2013;58:718–34.

- [13] Kim SM, Mudawar I. Universal approach to predicting two-phase frictional pressure drop for adiabatic and condensing mini/micro-channel flows. *Int J Heat Mass Transf* 2012;55:3246–61.
- [14] Kim SM, Mudawar I. Universal approach to predicting saturated flow boiling heat transfer in mini/micro-channels – part II. Two-phase heat transfer coefficient. *Int J Heat Mass Transf* 2013;64:1239–56.
- [15] Kim SM, Mudawar I. Universal approach to predicting heat transfer coefficient for condensing mini/micro-channel flow. *Int J Heat Mass Transf* 2013;56:238–50.
- [16] Kim SM, Mudawar I. Universal approach to predicting saturated flow boiling heat transfer in mini/micro-channels – part I. Dryout incipience quality. *Int J Heat Mass Transf* 2013;64:1226–38.
- [17] Kim SM, Mudawar I. Thermal design and operational limits of two-phase micro-channel heat sinks. *Int J Heat Mass Transf* 2017;106:861–76.
- [18] Kim SM, Mudawar I. Review of two-phase critical flow models and investigation of the relationship between choking, premature CHF, and CHF in micro-channel heat sinks. *Int J Heat Mass Transf* 2015;87:497–511.
- [19] Agrawal G, Kaisare NS, Pushpavanam S, Ramanathan K. Modelling the effect of flow mal-distribution on the performance of a catalytic converter. *Chem Eng Sci* 2012;71:310–20.
- [20] Jr Kitto JB, Robertson JM. Effects of maldistribution of flow on heat transfer equipment performance. *Heat Transf Eng* 1989;10:18–25.
- [21] Kim SM, Mudawar I. Review of databases and predictive methods for heat transfer in condensing and boiling mini/micro-channel flows. *Int J Heat Mass Transf* 2014;77:627–52.
- [22] Mueller AC, Chiou JP. Review of various types of flow maldistribution in heat exchangers. *Heat Transf Eng* 1988;9:36–50.
- [23] Xia GD, Jiang J, Wang J, Zhai YL, Ma DD. Effects of different geometric structures on fluid flow and heat transfer performance in microchannel heat sinks. *Int J Heat Mass Transf* 2015;80:439–47.
- [24] Anbumeenakshi C, Thansekhar MR. Experimental investigation of header shape and inlet configuration on flow maldistribution in microchannel. *Exp Therm Fluid Sci* 2016;75:156–61.
- [25] Kumaran RM, Kumaraguruparan G, Sornakumar T. Experimental and numerical studies of header design and inlet/outlet configurations on flow mal-distribution in parallel micro-channels. *Appl Therm Eng* 2013;58:205–16.
- [26] Tong JC, Sparrow EM, Abraham JP. Geometric strategies for attainment of identical outflows through all of the exit ports of a distribution manifold in a manifold system. *Appl Therm Eng* 2009;29:3552–60.
- [27] Chein R, Chen J. Numerical study of the inlet/outlet arrangement effect on microchannel heat sink performance. *Int J Therm Sci* 2009;48:1627–38.
- [28] Mishan Y, Mosyak A, Pogrebnyak E, Hetsroni G. Effect of developing flow and thermal regime on momentum and heat transfer in micro-scale heat sink. *Int J Heat Mass Transf* 2007;50:3100–14.
- [29] Said SAM, Ben-Mansour R, Habib MA, Siddiqui MU. Reducing the flow mal-distribution in a heat exchanger. *Comput Fluids* 2015;107:1–10.
- [30] Wei M, Fan Y, Luo L, Flamant G. CFD-based evolutionary algorithm for the realization of target fluid flow distribution among parallel channels. *Chem Eng Res Des* 2015;100:341–52.
- [31] Wei M, Boutin G, Fan Y, Luo L. Numerical and experimental investigation on the realization of target flow distribution among parallel mini-channels. *Chem Eng Res Des* 2016;113:74–84.
- [32] Ashrafi M, Kanani H, Shams M. Numerical and experimental study of two-phase flow uniformity in channels of parallel PEM fuel cells with modified Z-type flow-fields. *Energy* 2018;147:317–28.
- [33] Kumar R, Singh G, Mikieliewicz D. A new approach for the mitigating of flow maldistribution in parallel microchannel heat sink. *J Heat Transf* 2018;140.
- [34] Madane K, Kulkarni AA. Pressure equalization approach for flow uniformity in microreactor with parallel channels. *Chem Eng Sci* 2018;176:96–106.
- [35] Pistoresi C, Fan Y, Aubril J, Luo L. Fluid flow characteristics of a multi-scale fluidic network. *Chem Eng Process Process Intensif* 2018;123:67–81.
- [36] Yadav V, Kumar R, Narain A. Mitigation of flow maldistribution in parallel microchannel heat sink. *IEEE Trans Compon Packag Manuf Technol* 2019;9:247–61.
- [37] Kim S. A novel design method of the dividing header configuration using 3D numerical simulation for a heat exchanger with a parallel arrangement. *Appl Therm Eng* 2019;159:113807.
- [38] Kumar S, Singh PK. Effects of flow inlet angle on flow maldistribution and thermal performance of water cooled mini-channel heat sink. *Int J Therm Sci* 2019;138:504–11.
- [39] Dabrowski P. Thermohydraulic maldistribution reduction in mini heat exchangers. *Appl Therm Eng* 2020;173.
- [40] Song JY, Hah S, Kim D, Kim SM. Enhanced flow uniformity in parallel mini-channels with pin-finned inlet header. *Appl Therm Eng* 2019;152:718–33.
- [41] Gilmore N, Hassanzadeh-Barforoushi A, Timchenko V, Menicetas C. Manifold configurations for uniform flow via topology optimisation and flow visualisation. *Appl Therm Eng* 2021;183:116227.
- [42] Li Y, Roux S, Castelain C, Luo L, Fan Y. Tailoring the fluid flow distribution in a parallel mini-channel heat sink under multiple-peak heat flux. *Therm Sci Eng Prog* 2022;29.
- [43] Maharudrayya S, Jayanti S, Deshpande AP. Flow distribution and pressure drop in parallel-channel configuration of planar fuel cells. *J Power Sources* 2005;144:94–106.
- [44] Kim D, Yu CH, Yoon SH, Choi JS. Effects of manifold geometries on flow distribution to parallel microchannels. *J Mech Sci Tech* 2011;25:3069–74.
- [45] Habib MA, Ben-Mansour R, Said SAM, Al-Bagawi JJ, Al-Mansour KM. Correlations of flow maldistribution parameters in an air cooled heat exchanger. *Int J Numer Methods Fluids* 2008;56:143–65.
- [46] Saeed M, Kim MH. Numerical study on thermal hydraulic performance of water cooled mini-channel heat sinks. *Int J Refrig* 2016;69:147–64.
- [47] Liu X, Yu J. Numerical study on performances of mini-channel heat sinks with non-uniform inlets. *Appl Therm Eng* 2016;93:856–64.
- [48] Ansys Inc.. ANSYS Fluent Theory Guide (release 18.0). PA: Ansys Canonsburg; 2018.
- [49] Kader B. Temperature and concentration profiles in fully turbulent boundary layers. *Int J Heat Mass Transf* 1981;24:1541–4.
- [50] van Doormaal JP, Raithby GD. Enhancements of the SIMPLE method for predicting incompressible fluid flows. *Numer Heat Transf* 1984;7:147–63.
- [51] Shih TH, Liou WW, Shabbir A, Yang Z, Zhu J. A new k- ϵ eddy viscosity model for high reynolds number turbulent flows. *Comput Fluids* 1995;24:227–38.
- [52] E.W. Lemmon, M.L. Huber, M.O. McLinden, NIST standard reference database 23: Reference Fluid Thermodynamic and Transport Properties-REFPROP, Version 9.1, National Institute of Standards and Technology, Gaithersburg, MD, 2013.
- [53] Copeland D. Manifold microchannel heat sinks: analysis and optimization. *ASME JSME Therm Eng* 1995;4:169–74.
- [54] Shah RK, London AL. Laminar flow forced convection in ducts: a source book for compact heat exchanger analytical data. New York, NY: Academic Press; 1978.
- [55] Mohammadi M, Jovanovic GN, Sharp KV. Numerical study on flow uniformity and pressure characteristics within a microchannel array with triangular manifolds. *Comput Chem Eng* 2013;52:134–44.
- [56] Lu MC, Wang CC. Effect of the inlet location on the performance of parallel-channel cold-plate. *IEEE Trans Compon Packag Technol* 2006;29:30–8.
- [57] Kumaraguruparan G, Kumaran RM, Sornakumar T, Sundararajan T. A numerical and experimental investigation of flow maldistribution in a micro-channel heat sink. *Int Commun Heat Mass Transf* 2011;38:1349–53.
- [58] Pistoresi C, Fan Y, Luo L. Numerical study on the improvement of flow distribution uniformity among parallel mini-channels. *Chem Eng Process Process Intensif* 2015;95:63–71.
- [59] Wang CC, Yang KS, Tsai JS, Chen Y. Characteristics of flow distribution in compact parallel flow heat exchangers, part II: modified inlet header. *Appl Therm Eng* 2011;31:3235–42.
- [60] Solovitz SA, Mainka J. Manifold design for micro-channel cooling with uniform flow distribution. *J Fluids Eng* 2011;133:1–11.